

Introduction

The Problem: Representing Genomic Variation

- Pangenome graphs are the modern standard for representing structural variation in genomes, including complex, nested variants
- A central challenge: identifying bubbles, which, loosely defined, are subgraphs where paths diverge from a source vertex and reconverge at a sink

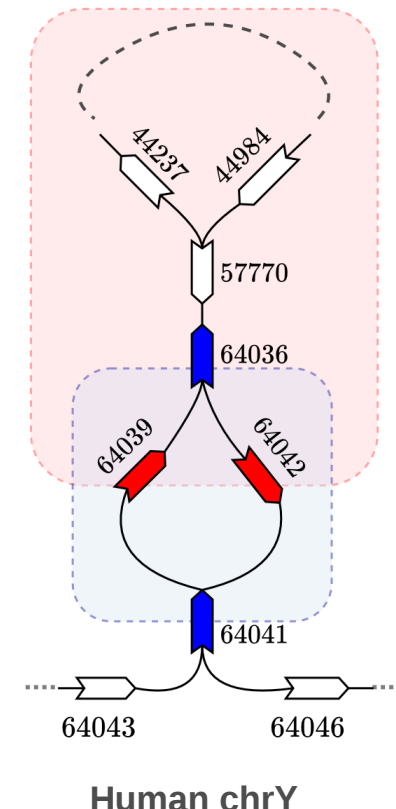
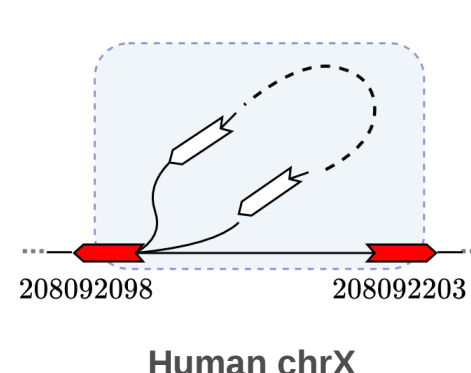
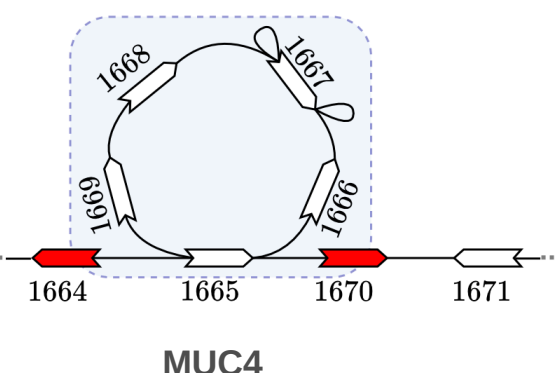
Why Existing Methods Fall Short

- Bubble detection is well-studied in directed graphs (in the context of genome assembly)
- But pangenome tools use bidirected graphs to represent double-stranded DNA.
- Prior bubble definitions adapted the superbubble concept to bidirected graphs. For example, snarls, bibubbles, and flubbles

Tool: <https://github.com/at-cg/billi>

Limitations of Prior Definitions

Bubble definition	Loops & cycles are allowed in subgraph	Bubbles do not partially overlap, nesting is allowed	Contiguity is ensured within subgraph for characterizing alleles
Ultrabubble	✗ (MUC4)	✓	✓
Superbubble	✗ (MUC4)	✓	✓
Snarls	✓	✗ (ChrY)	✗ (ChrX)
Compatible Snarls	✓	✓	✗ (ChrX)
Bibubble	✓	✗ (ChrY)	✓
Flubble	✓	✗ (ChrY)	✗ (ChrX)
Panbubble	✓	✓	✓

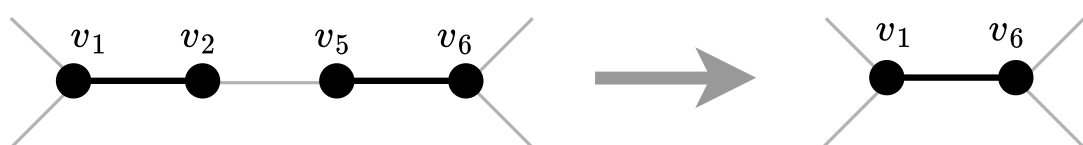


Notations

Biedged Graph $G_b = (V_b, E_b, f_b)$: An undirected colored multigraph where each edge is **black** or **gray**, every vertex has exactly one black edge, and at most one gray edge exists between any two vertices. $\hat{v}$ denotes the unique blackedge neighbor of v .

Walk: A vertex sequence $(v_0, v_1, \dots, v_n)$ that strictly alternates **black** and **gray** edges, starting and ending on black edges ($v_1 = \hat{v}_0, v_n = \hat{v}_{n-1}$).

Compaction Operation:



Definitions

Hairpin:

Given a compact biedged graph $G_b = (V_b, E_b, f_b)$, let the set of vertices reachable from vertex $s \in V_b$ without passing through s be denoted by $Y(s)$. A hairpin is said to exist at vertex s if

- every vertex $v \in Y(s)$ appears in an inverted closed walk starting from s ,
- removal of e_s , i.e., the blackedge connecting s and $\hat{s}$, separates the subgraph induced by $Y(s) \setminus \{s\}$ from the rest of the graph,
- no vertex in $Y(s)$ other than s satisfies these conditions.

The separable subgraph induced by $Y(s) \setminus \{s\}$ is called a hairpin and is denoted as $\langle s, t \rangle$

Panbubble:

A panbubble exists between two vertices s and t ($s \neq t$ and $s \neq \hat{t}$) in a compact biedged graph $G_b = (V_b, E_b, f_b)$ if all the following conditions hold:

matching: Let $U(s, t)$ denote the set of vertices reachable from s without passing through s or t . The matching condition requires that $U(s, t) = U(t, s)$.

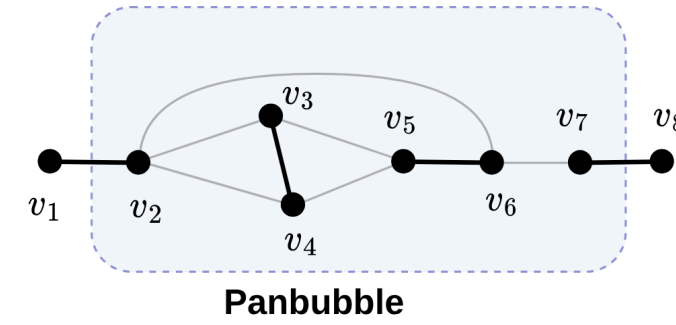
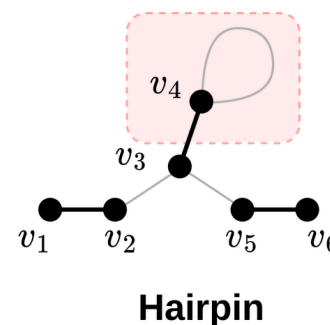
separable: Define $B(s, t)$ as $U(s, t) \setminus \{s, t\}$. If blackedges e_s and e_t are removed, the subgraph induced by $B(s, t)$ is disconnected from the rest of the graph.

contiguity: Every vertex $v \in U(s, t)$ appears on at least one walk from s to t .

no-hairpin: No vertex in $B(s, t) \setminus \{\hat{s}, \hat{t}\}$ is an entrance vertex of a hairpin.

minimality: No vertex in $U(s, t)$ other than t forms a pair with s and satisfies the above criteria. The panbubble is denoted as $\langle s, t \rangle$

Examples

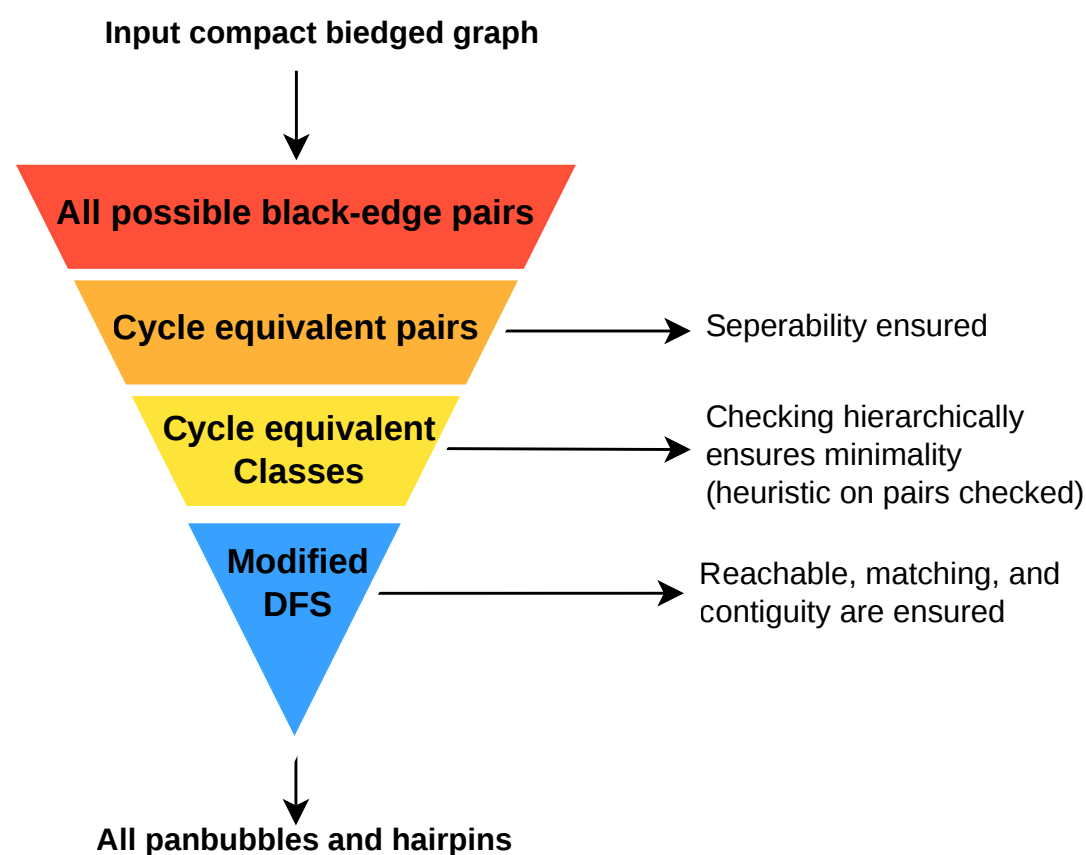


Problem Statment

Input: Compact biedged graph $G_b = (V_b, E_b, f_b)$

Output: (i) List of entrance vertex pairs of every panbubble in G_b
(ii) List of entrance vertices of every hairpin in G_b

Billi Algorithm Overview



Theorems

Theorem 1: Let $\langle x, y \rangle$ and $\langle m, n \rangle$ be two different panbubbles such that $B(x, y) \cap B(m, n) \neq \emptyset$, then either $B(x, y)$ is a strict subset of $B(m, n)$ or $B(m, n)$ is a strict subset of $B(x, y)$.

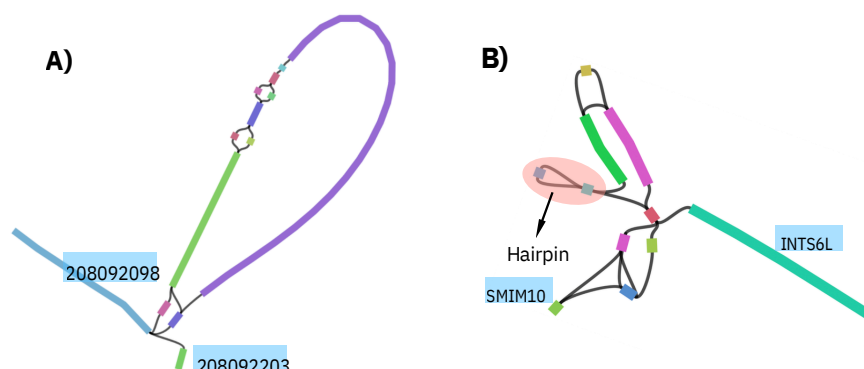
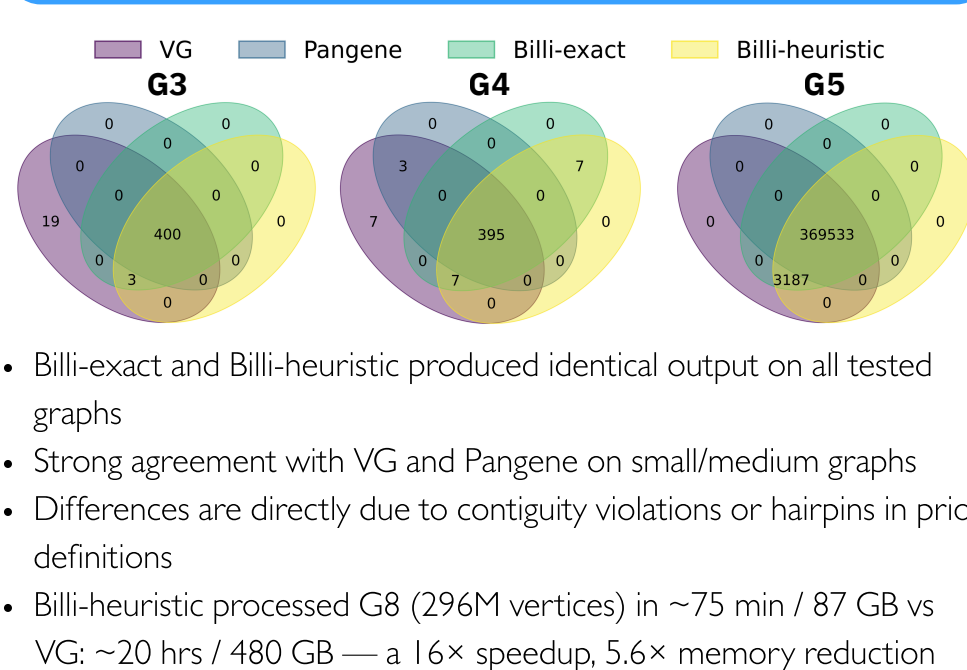
Theorem 2: Let $\langle m \rangle$ and $\langle n \rangle$ be two distinct hairpins in G_b . Then, the two hairpins do not share any vertex.

Experiments

- Evaluated on 8 publicly available pangenome graphs ranging from 36 to 296M vertices.
- Graph compaction is performed as a preprocessing step
- Billi (exact and heuristic) is compared against VG, which outputs snarls, and Pangene, which outputs bibubbles.

Graph	Source	Input Data
G1	minigraph	90 C4A/C4B haplotypes
G2	minigraph	57 MHC haplotypes
G3	pangene	50 E.coli genomes
G4	pangene	100 human + 10 great ape haplotypes
G5	HPRC v2 (G8)	Chr Y subgraph
G6	HPRC v2 (G8)	Chr X subgraph
G7	HPRC	Human Pangeneome v1 (94 haplotypes)
G8	HPRC	Human Pangeneome v2 (466 haplotypes)

Results



Conclusion

- Panbubbles and hairpins preserve contiguity, nesting, and no-hairpin guarantees that prior definitions lack
- Billi provides a practical, scalable foundation for variation discovery in large pangenome graphs

Future work

- Can panbubbles/hairpins be enumerated in linear time?
- Input biedged graph must have at least one tip. Can this be relaxed?
- Study how the panbubble count and structure evolve as haplotypes are added to a pangenome
- Integrate variant calling/genotyping pipelines for biological insights

References

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